

Avinash Chowdary Bonthu

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Summary

Software Engineer with 5+ years of experience designing and delivering scalable backend and cloud-native systems using Java, Spring Boot, and modern DevOps tools. Led end-to-end development of financial and healthcare platforms at Bank of America and Infosys, reducing deployment time and downtime by over 20% and managing systems handling over 1 million daily requests. Improved reliability by building fault-tolerant Kafka pipelines and optimizing real-time data processing across enterprise environments. Eager to leverage hands-on expertise in backend engineering and CI/CD automation to deliver robust, high-performance solutions.

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, C++, SQL, Shell Scripting
- **Frameworks & APIs:** Spring Boot, Spring MVC, Node.js, React, Flutter, FastAPI, Flask, REST APIs, GraphQL, gRPC
- **Python Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras, NLTK, Matplotlib
- **Databases & Cloud:** PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis, Kafka, AWS, Azure, GCP
- **DevOps & Tools:** Docker, Kubernetes, Helm, Git, GitLab CI/CD, Azure DevOps, Prometheus, Grafana, SonarQube, ArgoCD

EXPERIENCE

Bank of America

Mar 2025 - Present

Software Engineer

Remote, United States

- Owned end-to-end delivery of scalable backend services using Java and Spring Boot with RESTful APIs, supporting high-availability financial systems processing enterprise-scale transactions
- Spearheaded containerized deployments via Docker and Azure App Services, improving release consistency and reducing deployment cycle time by 25%
- Championed query optimization across PostgreSQL and MySQL for large relational datasets, improving data retrieval performance for real-time business reporting
- Orchestrated CI/CD pipelines using Azure DevOps, eliminating 25% of manual deployment effort across development and production environments
- Triaged and resolved production incidents using log analysis and system monitoring, reducing downtime by 20% for critical financial applications
- Integrated agentic, LLM-based tooling into backend workflows using prompt engineering and RAG over vector databases, accelerating incident triage and log analysis and improving developer productivity for critical financial services

Infosys

Jan 2022 - Jul 2024

Software Engineer

Hyderabad, TG, India

- Architected a cloud-native telemedicine platform using Java Spring Boot, PostgreSQL, and Kubernetes, supporting 160K+ users and 20K+ monthly consultations
- Spearheaded design and deployment of 20+ RESTful microservices using Spring Boot, Apache Kafka, and Elasticsearch, handling 1M+ daily requests with 25% lower latency and sub-second search over 50M+ records
- Pioneered a fault-tolerant real-time CDC pipeline using Kafka and Debezium, synchronizing 5+ mission-critical systems and slashing data inconsistency incidents by 60%
- Overhauled relational database queries using B-Tree indexing, reducing triaging time by 45% and contributing to \$1.5M business impact across telemedicine workflows
- Owned delivery of an enterprise healthcare workflow platform using Java backend services, onboarding 5K+ users across 10+ hospitals and reducing manual coordination by 70%
- Crafted analytics dashboards using Prometheus and Grafana, cutting incident detection time and reducing reporting turnaround by 40% with sub-5-second query latency
- Championed secure SDLC practices including input validation, authentication checks, and vulnerability scanning, achieving 35% faster vendor onboarding and 25% fewer transaction mismatches
- Mentored 10 interns through code reviews and testing standards, elevating code quality by 35% (SonarQube) and slashing onboarding time from 4 weeks to 2 weeks
- Engineered an NLP-based chatbot using RASA and supervised learning, achieving 92% intent classification accuracy across 20K+ real-world queries

- Launched a clinical scheduling system using Flutter and Spring Boot, cutting scheduling conflicts by 40% for a 500-bed hospital
- Built and productionized AI-powered recommendation systems and chatbots using LLMs, embeddings, and RAG over vector databases, guiding patients to the right clinics, doctors, and primary healthcare measures while improving access to timely, high-quality care

eBay

Oct 2020 - Dec 2021

Software Engineer

Remote, India

- Architected backend services using Java and RESTful APIs on AWS, powering high-volume e-commerce transactions and ensuring reliable distributed system performance
- Spearheaded scalable microservice design enabling seamless frontend-to-backend communication, improving modularity and maintainability across application architectures
- Owned SQL database interactions for efficient storage and retrieval of product and user data, supporting high-volume transactions and improving query performance
- Triaged application issues using logs and AWS monitoring tools, reducing defects by 20% and stabilizing platform performance for real-time transaction systems
- Orchestrated deployment and monitoring workflows using AWS services, improving response times by 15% and ensuring smooth production operations

EDUCATION

State University of New York at Buffalo

Aug 2024 - Jan 2026

Master of Science, Computer Science and Engineering

Buffalo, NY, US

- **Coursework:** Algorithms, Big Data Systems, Database Internals, Machine Learning, Functional Programming

Jaypee University of Engineering and Technology

Jul 2018 - May 2022

Bachelor of Technology, Computer Science and Engineering

Guna, MP, India

PROJECTS

TacoDB

- Architected a high-performance relational database engine with variable-length storage, buffer pool management, B-Tree indexing, and a custom query execution engine

Deep Learning Image Classifier

- Engineered and trained a CNN on 50K+ images, achieving 90% test accuracy and reducing overfitting by 25% via data augmentation and hyperparameter tuning

Distributed Network Chat App

- Spearheaded a TCP-based distributed chat system supporting real-time unicast and broadcast messaging with fault-tolerant communication and dynamic client management

ACHIEVEMENTS

- **Employee of the Month, Infosys:** Recognized for exceptional performance and outstanding contributions to telemedicine platform delivery
- **JPMC Software Virtual Internship:** Optimized a trading dashboard handling 10K+ real-time transactions/day, cutting data refresh latency by 25%
- **Sparks Foundation Graduate Intern:** Fine-tuned ML models on 50K+ financial records, achieving 85% classification accuracy and reducing prediction errors by 18%